

YONGE LI

AI/ML Engineer | AI Researcher | Data Scientist

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🌐 LinkedIn | 🏠 GitHub | 🌐 Portfolio

PROFESSIONAL SUMMARY

AI Engineering Masters Graduate with 15+ years of CNC machine tool industry experience. Strong background in Python, Data Science, Machine Learning, and Evolutionary Algorithm and optimization. Passionate about applying data-driven approaches to solve real-world challenges and continuously exploring emerging AI technologies.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, HTML, CSS

Machine Learning & AI: Machine Learning, Symbolic Regression, Evolutionary Computation (GE/GP/GA), Explainable AI (XAI) & Interpretable AI, Deep Learning, Reinforcement Learning (Q-Learning), Graph Neural Networks (GAT), Attention Mechanisms

Generative AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, LlamaIndex, Ollama, LangChain

Data Science & Analytics: Exploratory Data Analysis (EDA), Clustering (KMeans, GMM), Dimensionality Reduction (PCA, t-SNE), SVM, Decision Trees, Random Forest, XGBoost, Ridge/Lasso Regression, Cross-Validation, Statistical Analysis, Data Visualization, Data modeling, Predictive Modeling, Data Cleaning, Feature Engineering, Fairness-Aware ML

Knowledge Representation: RDF, SPARQL, Ontology Design, Knowledge Graphs

Frameworks & Libraries: Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Seaborn, PonyGE2, gplearn

Databases & Knowledge Systems: MySQL, SQL Server, MongoDB (No SQL), Snowflake, Vector Databases (Qdrant and Chroma), Knowledge Graphs, RDF, SPARQL

Tools & Platforms: Git, GitHub, SLURM (HPC), LaTeX, KNIME, Linux, VS Code

PROFESSIONAL EXPERIENCE

COMPtech AB, Jönköping

Oct 2025 – Jun 2026

AI Student Researcher – Master's Thesis

- Built an end-to-end machine learning pipeline for predicting mechanical properties of aluminium alloys from composition and processing data (313 records, 16 features), achieving $R^2 = 0.93$ on test set.
- Developed a hybrid symbolic regression framework (GE+Ridge) that produces interpretable equations competitive with XGBoost while enabling physical validation of predictions.
- Designed and automated a reproducible experiment pipeline on HPC (SLURM) with 30-seed replication across multiple configurations to ensure statistically reliable conclusions.
- Implemented data augmentation strategies calibrated to measurement uncertainties, improving model generalisation reliability by 50%.
- Benchmarked against five baselines (OLS, Ridge, Lasso, Random Forest, XGBoost); compared two symbolic-regression paradigms — Genetic Programming (gplearn) and Grammatical Evolution.
- A journal manuscript based on this work is currently in progress.

Technologies: Python, PonyGE2, gplearn, Scikit-learn, XGBoost, Ridge Regression, NumPy, Pandas, Matplotlib, SLURM (HPC), LaTeX, Git

Arfve / layr, Stockholm

Jun 2025 – Aug 2025

AI System Design Intern

- Designed the system architecture for a context-aware AI assistant across wearable platforms (earbuds, smart glasses, smartphones), defining a three-tier local processing framework with sensor fusion and action orchestration modules.
- Defined a dynamic multimodal input weighting scheme that adjusts voice, motion, and touch priorities based on environmental conditions, user activity states, and signal quality metrics.
- Proposed a tiered privacy-preserving architecture separating biometric processing (hardware enclave), personal preferences (edge), and generic queries (cloud) to minimise data exposure.
- Authored detailed use-case specifications for four deployment scenarios (healthcare, education, business, accessibility) with technical requirements and design trade-offs.

Technologies: System Architecture Design, Multi-Modal AI, Edge AI, Sensor Fusion, Privacy-Preserving Architecture, BLE/Wi-Fi Protocol Design

Hartford CNC Co., Ltd., Taiwan <i>Assistant Manager — Shanghai Representative Office</i>	Oct 2007 – Dec 2023
- Served as the primary liaison among customers, dealers, service engineers, and Taiwan headquarters for CNC machine-tool systems (Fanuc, Mitsubishi), handling both commercial operations and remote technical support. - Independently managed the Shanghai representative office, overseeing field service coordination, parts order processing, financial reporting, and government compliance as the sole administrative lead. <i>Technologies: CNC Systems (Fanuc, Mitsubishi), ERP Systems, Cross-functional Coordination</i>	

EDUCATION

Jönköping University , Jönköping, Sweden <i>Master of Science (M.Sc.) in Artificial Intelligence Engineering</i>	Aug 2024 – Jun 2026
Selected Academic Projects: ML Cost Prediction Model for Manufacturing: Built regression-based cost prediction models using Python. Logistics Analytics Project: Developed data analysis and machine learning models using Python and KNIME. RDF-Based Music Recommendation System: Built a recommendation system using RDF, SPARQL, and knowledge graphs. Slow Fashion Image Classification: Developed deep learning image classification models using python, CNN, and ResNet	
Heilongjiang University , Heilongjiang, China <i>Bachelor of Engineering (B.Eng.) in Electrical Engineering (Automation)</i>	Sep 2003 – Jun 2007

ADDITIONAL INFORMATION

Languages: Chinese (native) · English (professional) · Swedish (SFI Kurs C— ongoing)
References Available upon request. Professional references from industry and academic supervisors can be provided.